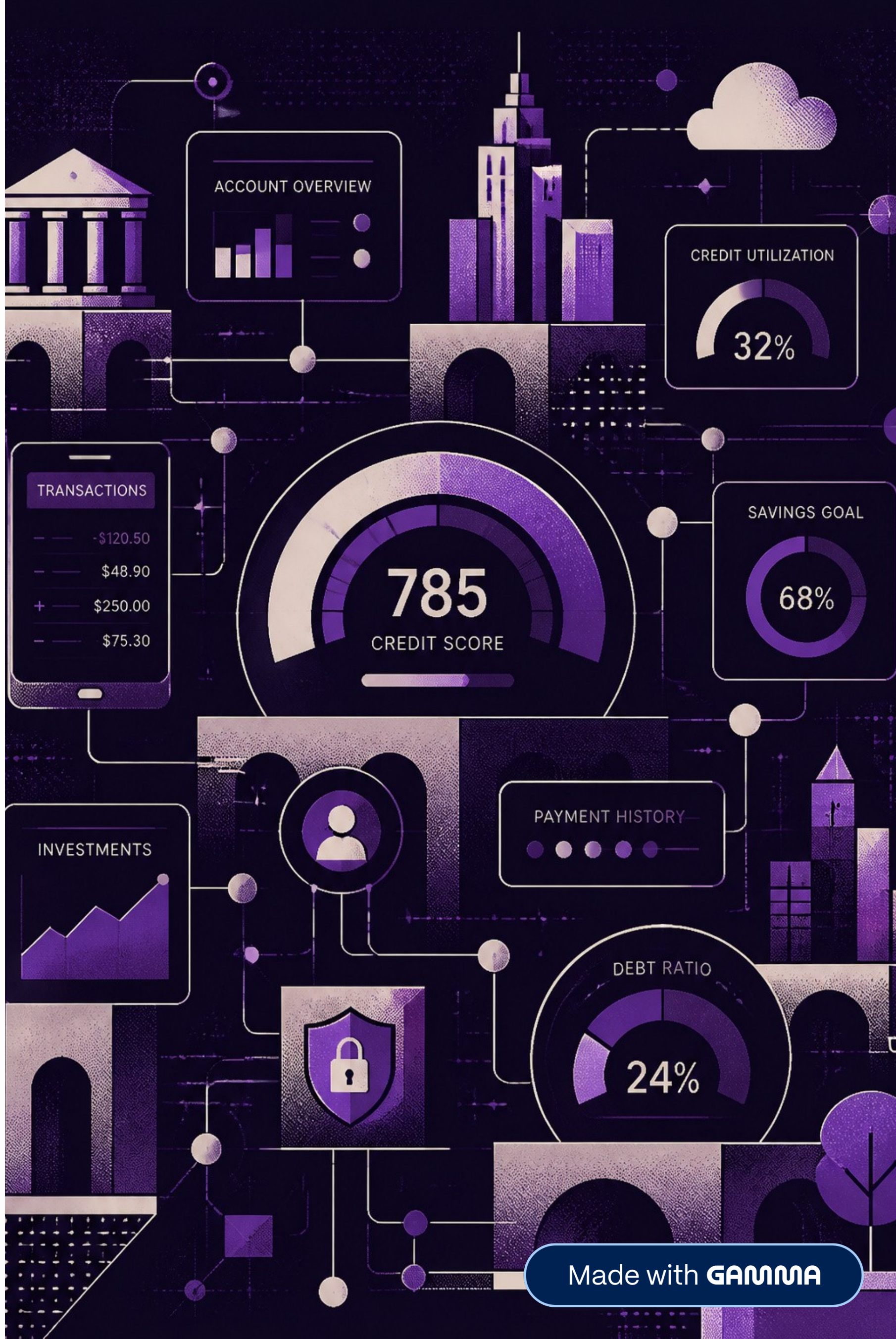
AI-Based Credit Risk Assessment System

Automating loan risk prediction with machine learning for faster, smarter, and more consistent lending decisions.



The Challenge: Manual Credit Evaluation

Time-Consuming Process

Manual review of each applicant is resource-intensive, creating bottlenecks and delays in loan approvals.

Human Error & Inconsistency

Subjective evaluations introduce bias and lead to inconsistent decisions across different reviewers.

Need for Data-Driven Solutions

Financial institutions urgently require automated, reliable systems to handle growing application volumes.

Project Objectives

Predict Credit Risk

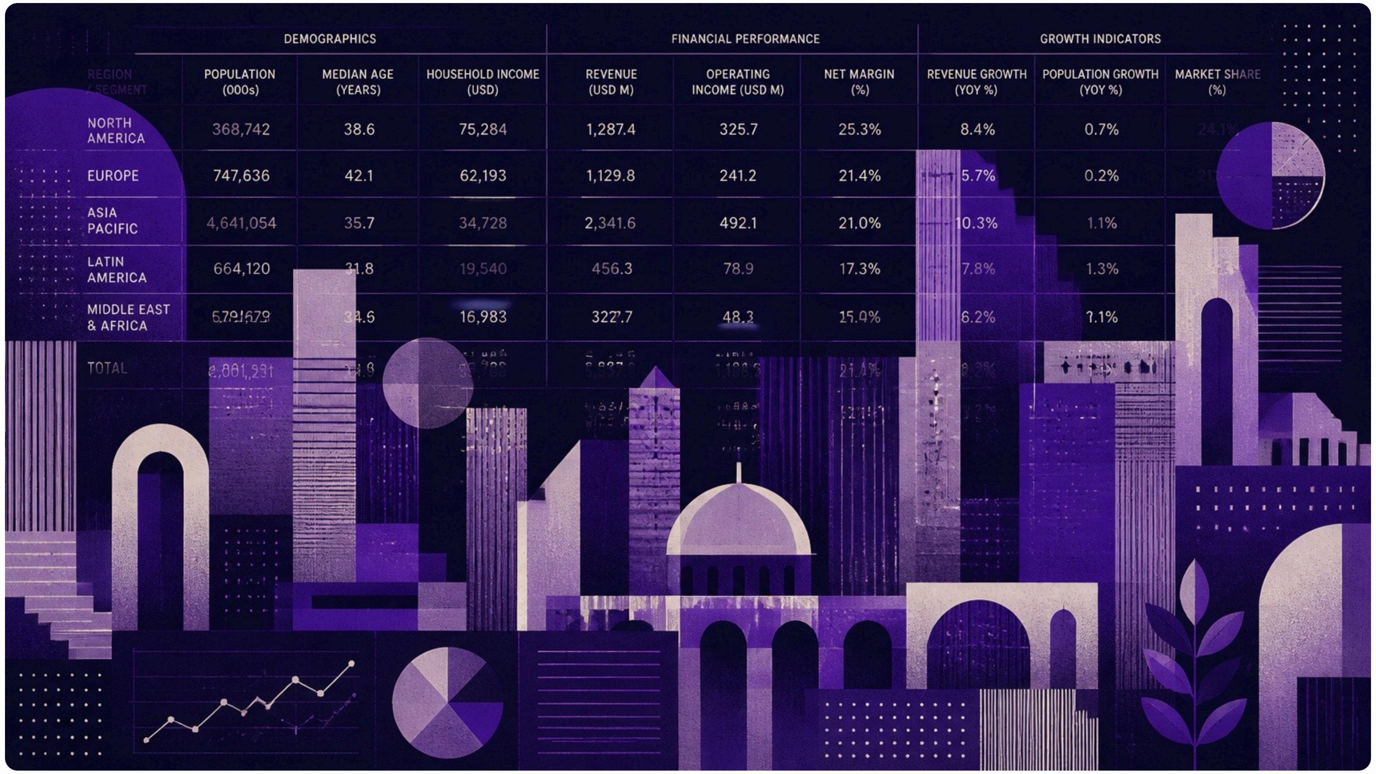
Develop an AI system to classify applicants as **Low** or **High** risk using machine learning.

Automate Loan Assessment

Replace manual preliminary evaluation with a fast, consistent, ML-powered pipeline.

Interactive Web Application

Deliver real-time predictions through a user-friendly interface accessible to non-technical staff.



Dataset & Features

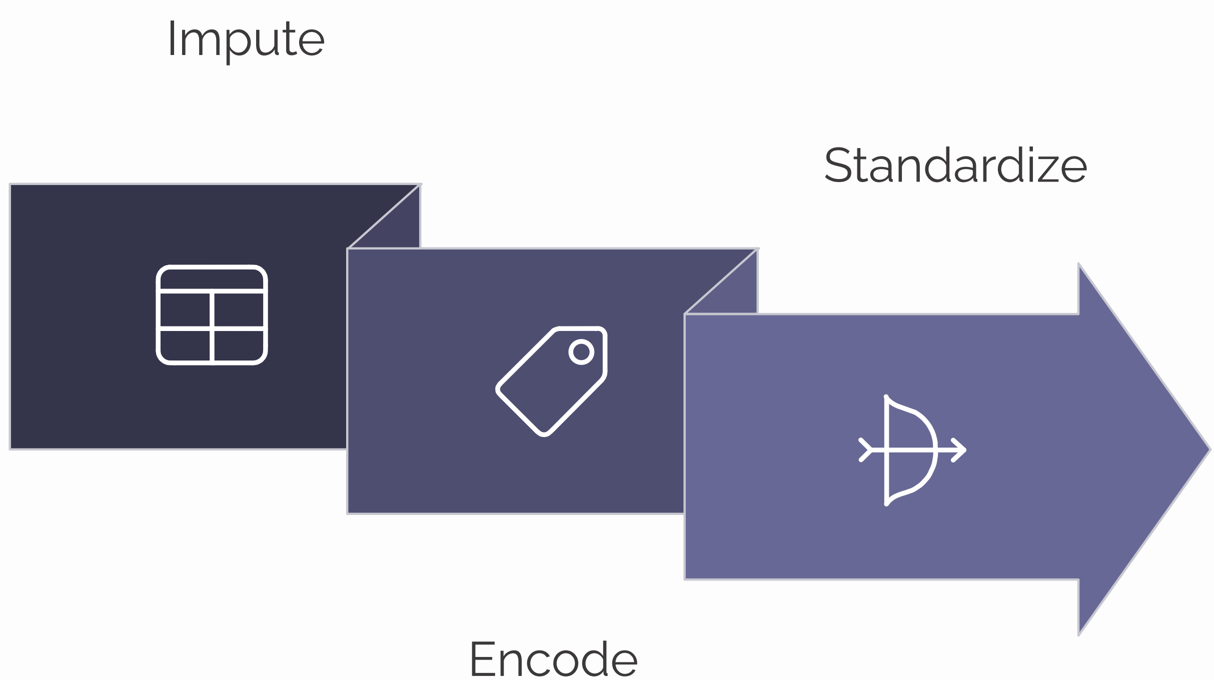
The model is trained on a **Loan Prediction dataset** containing demographic and financial attributes of applicants.

Key Variables

- Applicant income & co-applicant income
- Loan amount & repayment term
- Credit history & education level
- Employment status & property area

Target: Classify each applicant as **Low Risk** or **High Risk**

Data Preprocessing Pipeline



A clean, consistent preprocessing pipeline ensures the Random Forest model receives well-structured inputs, improving reliability and prediction accuracy.

Feature Engineering: Three Key Metrics

Total Income

Applicant Income + Co-applicant Income

Captures the full household earning capacity for repayment assessment.

Loan Income Ratio

Loan Amount ÷ Total Income

Measures debt burden relative to earnings — a core indicator of financial stress.

Estimated EMI

Loan Amount ÷ Repayment Term

Approximates monthly installment obligation to evaluate affordability.

Model Selection & Training

The system uses a **Random Forest Classifier** — an ensemble method that builds multiple decision trees and merges their outputs for robust, accurate predictions.

- Train-test split: **80% training / 20% testing**
- Test accuracy: **78.86%**
- Handles mixed data types and non-linear relationships well
- Resistant to overfitting through ensemble averaging

78.86%

Test Accuracy

Achieved on held-out 20% test set

80:20

Train-Test Split

Standard ML evaluation protocol

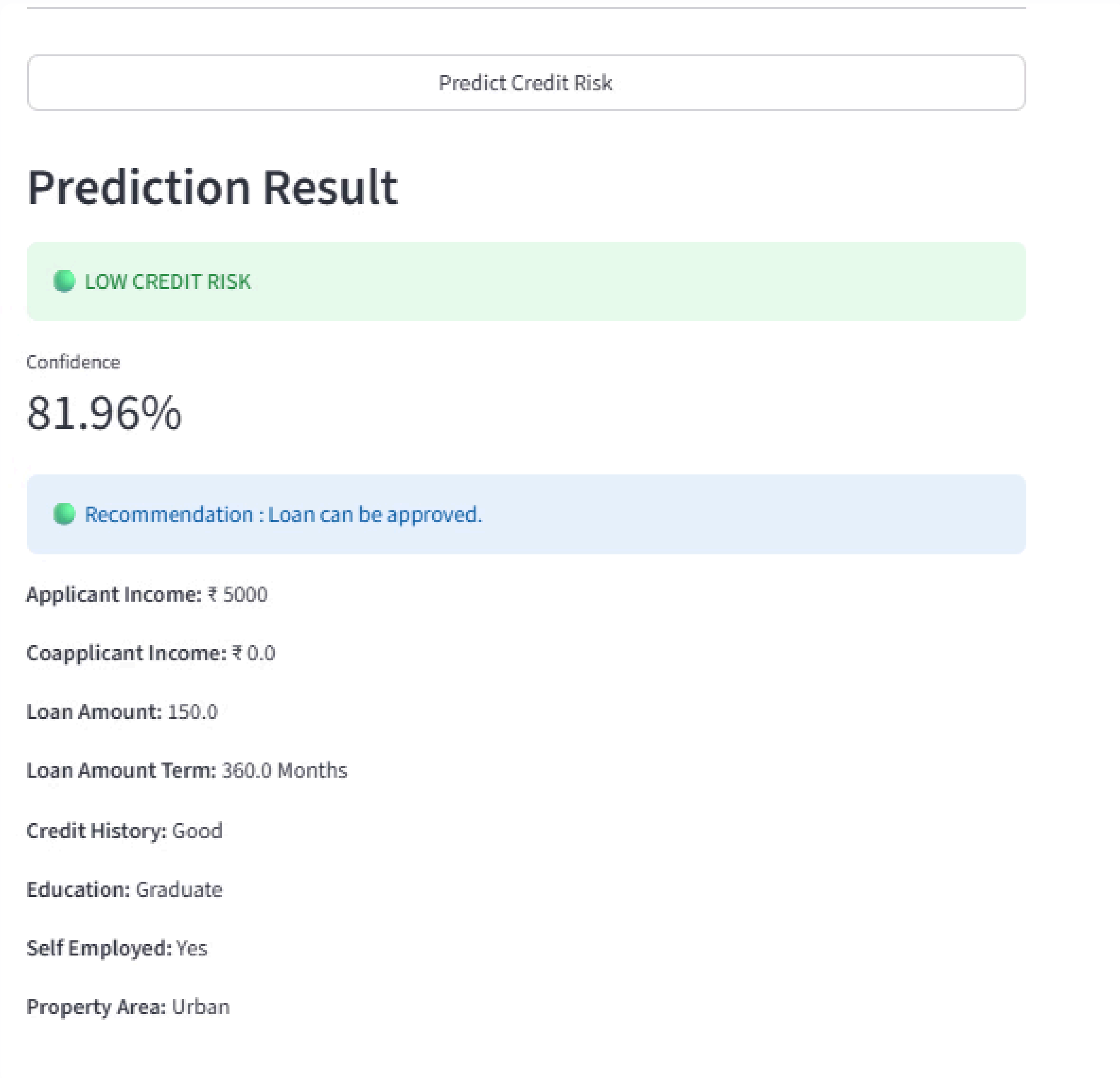
System Implementation

Interactive Streamlit Web Application

The system is deployed as a **Streamlit web app** providing real-time credit risk assessment through an intuitive interface.

- Input fields for all key applicant attributes
- Instant prediction with confidence score
- No coding knowledge required to operate
- Powered by a pre-trained, serialized Random Forest model

Application in Action: Low Risk Example



LOW RISK PREDICTION

Result: Low Credit Risk

The model classifies this applicant as **Low Credit Risk** with a confidence of **81.96%**.

Strong income indicators, positive credit history, and a favorable loan-income ratio all contribute to the high-confidence low-risk classification.

High confidence score signals a reliable, loan-approvable applicant profile.

Application in Action: High Risk Example

Predict Credit Risk

Prediction Result

❌ HIGH CREDIT RISK

Confidence

52.00%

🔴 Recommendation : Loan approval is risky.

Customer Summary

Applicant Income: ₹ 4000

Coapplicant Income: ₹ 0.0

Loan Amount: 150.0

Loan Amount Term: 360.0 Months

Credit History: Good

Education: Graduate

Self Employed: Yes

Property Area: Urban

HIGH RISK PREDICTION

Result: High Credit Risk

The model classifies this applicant as **High Credit Risk** with a confidence of **52%**.

Weaker financial indicators or an unfavorable credit history push this applicant into the high-risk category, flagging the case for further human review.

⚠ Lower confidence scores signal borderline cases — ideal for human-in-the-loop review.

Key Advantages

Fast Predictions

Delivers risk assessments within seconds, dramatically accelerating the loan approval workflow.

User-Friendly Interface

Designed for non-technical users — no coding or data science expertise required to operate.

Consistent & Unbiased

Algorithmic decision support eliminates human subjectivity for fairer, more consistent outcomes.

Scalable & Reusable

The trained model can be serialized, deployed at scale, and retrained as new data becomes available.